THU 5 NOV · 09:00–10:00 · PRIORY RD CMPLX F BLK 3F11

Introduction to Pure Mathematics

MATH10027 · Group 07 · Cardinality, the Pigeonhole Principle and permutations

THURSDAY 5 NOVEMBER 2026 · 09:00–10:00 · PRIORY RD CMPLX F BLK 3F11

Introduction to Pure Mathematics — lesson plan (55 min)

Cardinality, the Pigeonhole Principle and permutations.

Elapsed	What	Budget
00–04	Logistics (1 min) and the key-ideas page (3 min).	Start P1 at 04:00
04–09	P1 · Counting carefully: what exactly is an element? (Pure PS6 A1(a)–(d))	5 min
09–12	P2 · Finite or infinite? (oral round) (Pure PS6 A2(a)–(d))	3 min
12–18	P3 · The Pigeonhole Principle and the birthday-month party (Pure PS6 A4(a)(b) + A5)	6 min
18–25	P4 · Two people with the same number of friends (Pure PS6 B3)	7 min
25–31	P5 · Transitivity of $\leqslant$, and of strict inequality (Pure PS6 A6 + B1)	6 min
31–35	P6 · Inclusion–exclusion from disjoint pieces (Pure PS6 A7 (= Corollary 6.34))	4 min
35–41	P7 · A bijection from (0,1) onto the whole real line (Pure PS6 B4)	6 min
41–48	P8 · Composing permutations of {1,2,3}: order matters (Lecture 7.1.1 (3 Nov) + Pure PS6 B5 (partial))	7 min
48–52	P9 · Symmetries of a square as permutations of its vertices (Pure PS6 A8)	4 min
52–55	Exit check and reminders.	Finish 55:00

Opening words: “This week the lectures moved from functions to *sizes* of sets, and then to symmetries. Today: count carefully, use the Pigeonhole Principle properly, and compose permutations in the right order.”

> 00–01 先提醒 PS5 peer feedback 今天 13:00 截止（每人评两份）；01–04 用知识点页过三个定义：bijection = 同样大，injection = 至多，pigeonhole = 逆否。不要在开头讲群的定义——群还没在课上正式讲。

If the room is behind: If behind at minute 30: give P6 orally (one disjoint decomposition on the board) and go straight to P7. If behind at minute 45: do only gf and fg in P8 and skip P9 — the square is in the reserve bank and the student handout.

Each item: show the full question → say the idea → students write a first step → full boardwork → one-line check. The reserve bank below has answers for every other question on the sheet, for questions from the room.

00–01 · LOGISTICS

Today 13:00: PS5 peer feedback closes

Problem Sheet 5 peer feedback closes TODAY (Thursday 5 November) at 13:00.

1. On Blackboard, open Introduction to Pure Mathematics and go to the **PS5 submission point**.
2. Click **Start review**. You will be given **two anonymous submissions** to review.
3. For each submission write **one specific strength** and **one actionable improvement**, each tied to a question number (the peer-marked question is **Part B Q8**: preimages of unions, and $U \subseteq f^{-1}[f[U]]$ with equality when f is injective).
4. Submit both reviews before 13:00.

Good feedback names the question and the step, and says what to do next, not just what is wrong.

Model strength sentences

- B8(a): you proved both inclusions separately — first $f^{-1}[V_1 \cup V_2] \subseteq f^{-1}[V_1] \cup f^{-1}[V_2]$, then the reverse — and each started with "let $x \in \ldots$". Very easy to check.
- B8(b): you used the definition $f^{-1}[W] = \{x \in X : f(x) \in W\}$ explicitly, so it is clear why $u \in U$ gives $u \in f^{-1}[f[U]]$.

Model improvement sentences

- B8(a): only one inclusion is shown. To prove two sets are equal, also take $x \in f^{-1}[V_1] \cup f^{-1}[V_2]$ and show $f(x) \in V_1 \cup V_2$.
- B8(b): the equality part never uses injectivity. Take $x \in f^{-1}[f[U]]$: then $f(x) = f(u)$ for some $u \in U$, and injectivity gives $x = u \in U$ — write that step out.
- B8(b): please give an example showing the inclusion can be strict without injectivity, e.g. $f(x) = x^2$, $U = \{1\}$, $f^{-1}[f[U]] = \{-1, 1\}$.

Not useful: "Good work!", "Wrong.", "Needs more detail." — no question part and nothing to act on.

> 开场 1 分钟内讲完，不占主线 48 分钟。课程公告说只有约 30% 的同学做了互评，可以直说：大部分人还没做，今天 13:00 关。强调三点：具体、可操作、带题号；念一条好例句和一条坏例句（Good work!）对比即可。互评的是 PS5 Part B Q8（原像与并集、U ⊆ f⁻¹[f[U]]），学生可对照已发布的解答评。另：Quiz 6（计分，Part II 的 assessed homework）今天 13:00 开放、下周四 13:00 截止。学生问在哪里：Blackboard 课程页 → PS5 submission point → Start review，每人评两份匿名作业。

START HERE · 01–04 MINUTES

Key ideas: cardinality and permutations

Introduction to Pure Mathematics · Week 7 · Section 6.5 and Section 7.1 · Keep this page beside your working.

1 · Comparing sizes with functions

For a finite set, $|A|$ is the number of *distinct* elements. A set inside a set counts as one element: $\{1, \{1, 2\}\}$ has two elements.

Same size. $|A| = |B|$ means there is a **bijection** $A \to B$.
At most. $|A| \leqslant |B|$ means there is an **injection** $A \to B$ (Definition 6.31; for finite sets this agrees with counting, Proposition 6.30). $|A| < |B|$ means $|A| \leqslant |B|$ and $|A| \neq |B|$.

If $A \subseteq B$, the inclusion $a \mapsto a$ is injective, so $|A| \leqslant |B|$. For infinite sets a proper subset can have the same size: $n \mapsto 2n$ is a bijection $\mathbb{Z} \to 2\mathbb{Z}$.

Schröder–Bernstein (Theorem 6.32). If $|A| \leqslant |B|$ and $|B| \leqslant |A|$, then $|A| = |B|$. You may use it; you do not need its proof.

2 · Counting rules for finite sets

$$A \cap B = \varnothing \implies |A \cup B| = |A| + |B| \quad (\text{Prop. } 6.33), \qquad |A \cup B| = |A| + |B| - |A \cap B| \quad (\text{Cor. } 6.34).$$

Pigeonhole Principle. If $|A| > |B|$, then no function $f : A \to B$ is injective: some $a_1 \neq a_2$ have $f(a_1) = f(a_2)$. It is the contrapositive of "injection $\Rightarrow |A| \leqslant |B|$". To use it, say clearly *what the pigeons are, what the holes are, and what the function is*.

3 · Permutations (Section 7.1)

A **permutation** of X is a bijection $X \to X$ (Definition 7.1). The six permutations of $\{1, 2, 3\}$, with the lecture labels:

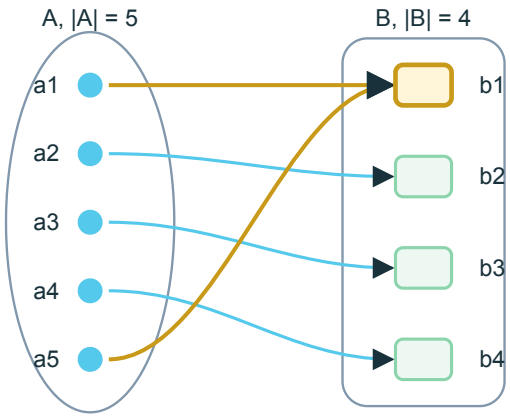
x	e	f	g	h	i	j
$1 \mapsto$	1	1	2	3	2	3
$2 \mapsto$	2	3	1	1	3	2
$3 \mapsto$	3	2	3	2	1	1

Composition is read right to left: $fg = f \circ g$ means "do g first, then f". For example $gf = i$ but $fg = h$, so the order matters. Composites of bijections are bijections (Corollary 6.19), and every permutation has an inverse permutation.

Proof checklist. Same size → write down a bijection and prove it is one (or give injections both ways and quote Theorem 6.32). Pigeonhole → define the function and compare sizes. Finite counting → split into *disjoint* pieces first.

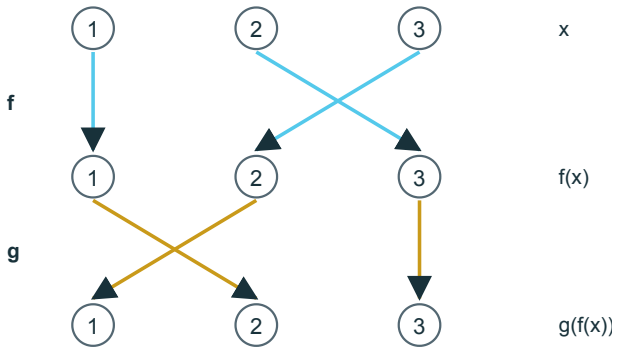
VISUAL · USE DURING P3–P4 AND P8

Pigeonhole and composing permutations



More inputs than boxes ⇒ no injection (Pigeonhole)

5 > 4: whatever the arrows are, two inputs must share a box. Contrapositive of Prop 6.30.



gf = g∘f = i (right-to-left: f first)

gf = i, but fg = h: the order of composition matters.

Main route · 04–52

9 items, 48 minutes including student writing time.

MAIN ROUTE · 04–09 elapsed · 5 min

P1 · Counting carefully: what exactly is an element?

Pure PS6 A1(a)–(d)

A1. Calculate the cardinality of the following finite sets:

(a) $A = \{1, 2, \{1, 2\}, 12\}$

(b) $B = \{x \in \mathbb{Z}_+ : 1 \leqslant x \leqslant 30,\ \gcd(x, 30) = 1\}$

(c) $C = \mathrm{im}(f)$, where $f\colon X \to \mathbb{Z}$ with $f(x) = x^2$ and $X = \{x \in \mathbb{Z} : -5 \leqslant x \leqslant 3\}$

(d) $D = \bigcap_{n \in \mathbb{Z}_+} (-1/n,\ 2/n)$, where $(-1/n,\ 2/n) = \{x \in \mathbb{R} : -1/n < x < 2/n\}$

> **Say:** The cardinality of a finite set is just the number of different elements, but you have to be honest about what the elements are. In (a), is $\{1, 2\}$ one element or two? In (d), ask yourself: which real numbers survive in every single interval?

> 板书先写 (a)：把四个元素逐个圈出来，强调 {1,2} 是一个元素、12 是十二。常见错误：(a) 答 3 或 5；(d) 答 (−1,2) 或空集。追问：0 在不在每个区间里？x = 0.001 在不在 n = 10000 的区间里？(d) 必须点名 Archimedean Property (Thm 5.13)，正负两种情况都写。时间紧：(b)(c) 只报答案加一句理由，(d) 写完整。

Students write a first step (30–60 s) → then board.

(a) The elements of A are 1, 2, $\{1, 2\}$ and 12. These four objects are different: $\{1, 2\}$ is a set, not a number, so it is a single element different from 1, 2 and 12; and 12 is the integer twelve, not the digits 1 and 2. Hence $|A| = 4$.

(Note: $1 \in A$ and $\{1, 2\} \in A$, and also $\{1, 2\} \subseteq A$. Only the elements are counted.)

(b) Since $30 = 2 \cdot 3 \cdot 5$, we have $\gcd(x, 30) = 1$ if and only if none of $2, 3, 5$ divides x. Listing $1, \ldots, 30$:

$$B = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad |B| = 8.$$

(Check: remove the 15 even numbers, then the 5 odd multiples of 3, namely $3, 9, 15, 21, 27$, then the 2 remaining multiples of 5, namely $5, 25$: $30 - 15 - 5 - 2 = 8$.)

(c) $X = \{-5, -4, \ldots, 3\}$ has 9 elements, and

$$C = \{25, 16, 9, 4, 1, 0, 1, 4, 9\} = \{0, 1, 4, 9, 16, 25\}, \qquad |C| = 6.$$

(Repeated values count once: f is not injective, e.g. $f(-3) = f(3)$, so $|\mathrm{im}(f)| < |X|$.)

(d) Claim: $D = \{0\}$, so $|D| = 1$.

- $0 \in D$: for every $n \in \mathbb{Z}_+$ we have $-1/n < 0 < 2/n$.
- Let $x > 0$. By the Archimedean Property (Theorem 5.13) there is $n \in \mathbb{Z}_+$ with $n > 2/x$. Then $2/n < x$, so $x \notin (-1/n, 2/n)$, and hence $x \notin D$.

- Let $x < 0$. Then $-x > 0$, so by the Archimedean Property there is $n \in \mathbb{Z}_+$ with $n > 1/(-x)$. Then $1/n < -x$, i.e. $x < -1/n$, so $x \notin (-1/n, 2/n)$, and hence $x \notin D$.

Therefore $D = \{0\}$ and $|D| = 1$. Every interval in the intersection is infinite, but only 0 lies in all of them.

> **Check:** What is $|\{\emptyset, \{\emptyset\}, \{1, 2\}, \{2, 1\}\}|$?
> **Answer:** 3: $\{1, 2\} = \{2, 1\}$ is the same element, while $\emptyset \neq \{\emptyset\}$.

MAIN ROUTE · 09–12 elapsed · 3 min

P2 · Finite or infinite? (oral round)

Pure PS6 A2(a)–(d)

A2. For each of the following, determine if the given set is finite:

(a) $A = \{x \in \mathbb{Z}_+ : \gcd(x, 30) = 1\}$

(b) $B = \{x \in \mathbb{Q} : \exists\, y \in \mathbb{Q},\ xy = 2\}$

(c) $C = \mathrm{im}(f) \cap \mathbb{Z}$, where $f\colon \mathbb{R} \to \mathbb{R}$ is the sine function $f(x) = \sin(x)$

(d) $D = \bigcup_{n \in \mathbb{Z}_+} (-1/n,\ 2/n)$, where $(-1/n,\ 2/n) = \{x \in \mathbb{R} : -1/n < x < 2/n\}$

> **Say:** Quick oral round: for each set say finite or infinite, with a one-line reason. To show a set is infinite, it is enough to find infinitely many different elements inside it.

> 口答为主，每小题 30–40 秒，点学生说结论，教师板书一句理由。(b) 学生常漏掉 0 不在 B 里；(c) 常误答无限（把 im(f) 和 im(f)∩Z 混淆）；(d) 与 A1(d) 并排对照：交集 {0}，并集 (−1,2)。时间紧：只写 (b)(d) 的理由，(a)(c) 口头。

Students write a first step (30–60 s) → then board.

(a) Infinite. Let $n \in \mathbb{Z}_+$. Any common divisor d of $30n + 1$ and 30 divides $(30n + 1) - n \cdot 30 = 1$, so $\gcd(30n + 1, 30) = 1$ and $30n + 1 \in A$. The numbers $31, 61, 91, \ldots$ are all different, so A contains infinitely many elements.

(b) Infinite. If $x = 0$ then $xy = 0 \neq 2$ for every y, so $0 \notin B$. If $x \in \mathbb{Q}$ and $x \neq 0$, then $y = 2/x \in \mathbb{Q}$ satisfies $xy = 2$, so $x \in B$. Hence $B = \mathbb{Q} \setminus \{0\}$, which contains $\mathbb{Z}_+$, so B is infinite.

(c) Finite. Since $-1 \leqslant \sin x \leqslant 1$ for all x, we get $C \subseteq [-1, 1] \cap \mathbb{Z} = \{-1, 0, 1\}$. All three values occur: $\sin(-\pi/2) = -1$, $\sin 0 = 0$, $\sin(\pi/2) = 1$. So $C = \{-1, 0, 1\}$ and $|C| = 3$.

(d) Infinite. For every $n \in \mathbb{Z}_+$ we have $-1 \leqslant -1/n$ and $2/n \leqslant 2$, so $(-1/n, 2/n) \subseteq (-1, 2)$, and the interval for $n = 1$ is $(-1, 2)$ itself. Hence $D = (-1, 2)$. It contains the different numbers $1/k$ for all $k \in \mathbb{Z}_+$, so D is infinite.

Compare with A1(d): the intersection of the same intervals is $\{0\}$ (finite); the union is $(-1, 2)$ (infinite).

> **Check:** Is $\{x \in \mathbb{Z} : \exists\, y \in \mathbb{Z},\ xy = 2\}$ finite?
> **Answer:** Yes: it is $\{-2, -1, 1, 2\}$. Compare (b), where y may be any rational number.

MAIN ROUTE · 12–18 elapsed · 6 min

P3 · The Pigeonhole Principle and the birthday-month party

Pure PS6 A4(a)(b) + A5

A4. Let A and B be finite sets with $|A| = m$ and $|B| = n$.

(a) Prove that there is no surjective function $f\colon A \to B$ if $m < n$.

(b) Prove that there is no injective function $f\colon A \to B$ if $m > n$.

A5. What is the smallest number of people you need to invite to a party in order to guarantee that at least two of your invited guests have birthdays in the same month of the year?

> **Say:** Part (b) is the Pigeonhole Principle written with functions: if A has more elements than B, no function $A \to B$ can be injective. It is just the contrapositive of Proposition 6.30. For the party, a smallest-number answer always has two halves: why 13 works, and why 12 does not.

> 板书先写 Prop 6.30 的一个方向：存在单射 ⇒ |A| ≤ |B|，正下方写它的逆否命题，标注：这就是 Pigeonhole = A4(b)。A5 学生常只答 13 不说明 12 为什么不够；强调最小两个字要证两件事。追问：要保证至少三人同月需要多少人？（25）时间紧：A4(a) 口头一句像集最多 m 个元素，A4(b) 和 A5 写完整。

Students write a first step (30–60 s) → then board.

Set-up. Write $A = \{a_1, \ldots, a_m\}$ with $a_1, \ldots, a_m$ different. For any function $f\colon A \to B$, $\mathrm{im}(f) = \{f(a_1), \ldots, f(a_m)\}$.

A4(a). The list $f(a_1), \ldots, f(a_m)$ has m entries, so $\mathrm{im}(f)$ has at most m elements:

$$|\mathrm{im}(f)| \leqslant m < n = |B|.$$

Hence $\mathrm{im}(f) \neq B$: some $b \in B$ is not of the form $f(a)$. So f is not surjective. Since f was arbitrary, there is no surjection $A \to B$.

A4(b). Suppose $f\colon A \to B$ is injective. If $f(a_k) = f(a_l)$ then $a_k = a_l$, so $k = l$. Thus $f(a_1), \ldots, f(a_m)$ are m different elements of B, and so $m \leqslant n$. We have shown

$$f \text{ injective} \implies m \leqslant n,$$

and its contrapositive is: if $m > n$ then f is not injective. As f was arbitrary, no injection $A \to B$ exists.

Link with lectures. Proposition 6.30 says: for non-empty finite sets, $|A| \leqslant |B|$ if and only if there is an injective $f\colon A \to B$. The direction "there is an injection $\Rightarrow |A| \leqslant |B|$" has contrapositive

$$|A| > |B| \implies \text{no function } A \to B \text{ is injective},$$

which is exactly A4(b): the **Pigeonhole Principle**. For every $f\colon A \to B$ there are $a \neq a'$ in A with $f(a) = f(a')$.

A5. Answer: 13.

- *13 guests are enough.* Let G be a set of 13 guests, M the set of 12 months, and $b\colon G \to M$ send each guest to their birthday month. Since $|G| = 13 > 12 = |M|$, b is not injective by A4(b): there are two different guests $x \neq y$ with $b(x) = b(y)$.

- *12 guests are not enough.* Twelve guests could have birthdays in January, February, ..., December, one in each month; then no two share a month. (With fewer guests, take some of these.) So no number below 13 guarantees a shared month.

Hence the smallest number is 13.

> **Check:** How many guests guarantee that at least three of them share a birthday month?
> **Answer:** 25: with 24 guests, two per month is possible; with 25 guests, if each month had at most two there would be at most 24 guests.

MAIN ROUTE · 18–25 elapsed · 7 min

P4 · Two people with the same number of friends

Pure PS6 B3

B3. Let $n \geqslant 2$ be an integer. Show that in any group of n people, two of them will have the same number of friends within this group. (You may assume that if person A is friends with person B, then B is also friends with A.)

> **Say:** Make a function: each person goes to their number of friends. There are n people and n possible values $0, 1, \ldots, n-1$, so the pigeonhole does not bite yet. The trick: can one person have 0 friends while another has $n-1$?

> 板书先写函数 F: X → {0,…,n−1}，旁边写一句：|X| = n = 值域候选个数，还不能直接用 pigeonhole。核心一句：0 和 n−1 不能同时出现（有 n−1 个朋友的人和所有人都是朋友，包括那个 0 朋友的人，矛盾），这里要用到友谊的对称性。常见错误：直接说 n 个人只有 n−1 个值，却不解释为什么；或分情况时把两种情况写成同一种。追问：n = 2 时是什么情形？（两人要么都是 0，要么都是 1）时间紧：Case 2 说同理即可。

Students write a first step (30–60 s) → then board.

Label the people $X = \{1, 2, \ldots, n\}$. For $k \in X$ let $F(k)$ be the number of friends of person k within the group. Nobody is their own friend, so each person has at most $n-1$ friends, and

$$F\colon X \to \{0, 1, \ldots, n-1\}.$$

We must show that F is not injective. Note that X and $\{0, 1, \ldots, n-1\}$ both have n elements, so the Pigeonhole Principle cannot be applied straight away.

Key observation: the values 0 and $n-1$ cannot both occur. Suppose $F(k) = n-1$. Then person k is friends with every other person l, so (by symmetry of friendship) every $l \neq k$ is friends with k, giving $F(l) \geqslant 1$. Also $F(k) = n-1 \geqslant 1$. So nobody has 0 friends.

Case 1: someone has 0 friends. By the observation nobody has $n-1$ friends, so $F(k) \in \{0, 1, \ldots, n-2\}$ for all k. This set has $n-1$ elements and $|X| = n > n-1$, so by the Pigeonhole Principle (A4(b)) F is not injective.

Case 2: nobody has 0 friends. Then $F(k) \in \{1, 2, \ldots, n-1\}$ for all k, again a set with $n-1$ elements, so F is not injective by A4(b).

In both cases there are two different people $k \neq l$ with $F(k) = F(l)$, i.e. with the same number of friends. (The condition $n \geqslant 2$ is needed so that the group contains two people at all.)

> **Check:** Five people: could their numbers of friends be exactly $0, 1, 2, 3, 4$?
> **Answer:** No: the person with 4 friends is friends with everyone, including the person with 0 friends, which is a contradiction.

MAIN ROUTE · 25–31 elapsed · 6 min

P5 · Transitivity of ⩽, and of strict inequality

Pure PS6 A6 + B1

Recall (Definition 6.31). For any sets A, B we write $|A| \leqslant |B|$ if there exists an injective function $A \to B$, and $|A| < |B|$ if $|A| \leqslant |B|$ and $|A| \neq |B|$. *Schröder–Bernstein Theorem (Theorem 6.32):* if $|A| \leqslant |B|$ and $|B| \leqslant |A|$, then $|A| = |B|$.

A6. Let A, B and C be sets such that $|A| \leqslant |B|$ and $|B| \leqslant |C|$. Prove that $|A| \leqslant |C|$.

B1. Let A, B and C be sets such that $|A| < |B|$ and $|B| < |C|$. Prove that $|A| < |C|$.

> **Say:** A6 is just: compose the two injections. In B1 the hard half is not equal: for infinite sets you cannot count, and $|A| \neq |C|$ means no bijection exists at all, so we argue by contradiction using Schröder–Bernstein.

先把 Definition 6.31（⩽ 和严格小于的定义）写在黑板左上角，这是本题全部原料。A6 两行写完；B1 分两步：⩽ 来自 A6，≠ 用反证——假设 |A| = |C|，用 C→A 的双射接上 B→C 的单射得到 B→A 的单射，再用 Schröder–Bernstein (Thm 6.32) 得 |A| = |B|，矛盾。常见错误：说 g∘f 不是满射，所以 |A| ≠ |C|——立刻用 2Z ⊆ Z 的反例打掉。官方解答 A6 引用的是 Prop 6.30（只对有限集），本题是任意集合，应引用 Def 6.31。时间紧：A6 口述，B1 写完整。

Students write a first step (30–60 s) → then board.

A6. Since $|A| \leqslant |B|$, by Definition 6.31 there is an injective $f\colon A \to B$; since $|B| \leqslant |C|$ there is an injective $g\colon B \to C$. Consider $g \circ f\colon A \to C$. Let $a_1, a_2 \in A$ with $(g \circ f)(a_1) = (g \circ f)(a_2)$, i.e. $g(f(a_1)) = g(f(a_2))$. As g is injective, $f(a_1) = f(a_2)$; as f is injective, $a_1 = a_2$. So $g \circ f$ is injective (this is Theorem 6.18(a)), and therefore $|A| \leqslant |C|$.

B1. *Step 1.* From $|A| < |B|$ and $|B| < |C|$ we have in particular $|A| \leqslant |B|$ and $|B| \leqslant |C|$, so $|A| \leqslant |C|$ by A6.

Step 2: $|A| \neq |C|$. Suppose, for a contradiction, that $|A| = |C|$. Then there is a bijection $A \to C$; its inverse $k\colon C \to A$ is a bijection, in particular injective. Let $g\colon B \to C$ be injective (from $|B| \leqslant |C|$). Then $k \circ g\colon B \to A$ is injective (composition of injections, as in A6), so $|B| \leqslant |A|$. Together with $|A| \leqslant |B|$, the Schröder–Bernstein Theorem (Theorem 6.32) gives $|A| = |B|$, contradicting $|A| < |B|$. Hence $|A| \neq |C|$.

By Steps 1 and 2, $|A| \leqslant |C|$ and $|A| \neq |C|$, that is, $|A| < |C|$.

Why Step 2 is not automatic. For infinite sets, $|A| \neq |C|$ means that *no* bijection $A \to C$ exists at all. Knowing that the particular injection $g \circ f$ is not surjective proves nothing: the inclusion $2\mathbb{Z} \to \mathbb{Z}$, $x \mapsto x$, is injective and not surjective, yet $x \mapsto 2x$ is a bijection $\mathbb{Z} \to 2\mathbb{Z}$, so $|2\mathbb{Z}| = |\mathbb{Z}|$. We cannot just count, so we argue by contradiction with Schröder–Bernstein. (For finite sets it is automatic: $m < n < p$ gives $m < p$.)

> **Check:** True or false: if $f\colon A \to B$ is injective but not surjective, then $|A| < |B|$.
> **Answer:** False in general: the inclusion $2\mathbb{Z} \to \mathbb{Z}$ is injective and not surjective, but $|2\mathbb{Z}| = |\mathbb{Z}|$. (True for finite sets.)

MAIN ROUTE · 31–35 elapsed · 4 min

P6 · Inclusion–exclusion from disjoint pieces

Pure PS6 A7 (= Corollary 6.34)

A7. Let A and B be non-empty finite sets. Prove that $|A \cup B| = |A| + |B| - |A \cap B|$.

[You may use Proposition 6.33: if A and B are disjoint finite subsets of a set X, then $|A \cup B| = |A| + |B|$.]

> **Say:** Proposition 6.33 only works for pieces that do not overlap, so cut $A \cup B$ into A and the part of B outside A. Then cut B itself the same way.

> 板书画一个两圆 Venn 图，用两种颜色标出 A∪B = A ⊔ (B∖A) 和 B = (B∖A) ⊔ (A∩B)，每条等式旁写 disjoint ⇒ Prop 6.33。常见错误：只写交集被数了两次所以减掉——那是直觉不是证明，必须化归到 Prop 6.33。追问：三个集合的版本是什么？时间紧：两个集合等式（Step 1、2 的集合相等）只写不证。

Students write a first step (30–60 s) → then board.

All sets below are finite subsets of $X = A \cup B$.

Step 1. $A \cup B = A \cup (B \setminus A)$, and $A \cap (B \setminus A) = \emptyset$.

Indeed, $\supseteq$ holds because $B \setminus A \subseteq B$. For $\subseteq$: let $x \in A \cup B$; if $x \in A$ we are done, and if $x \notin A$ then $x \in B$, so $x \in B \setminus A$. The two pieces are disjoint since no element is both in A and not in A. By Proposition 6.33,

$$|A \cup B| = |A| + |B \setminus A|. \qquad (1)$$

Step 2. $B = (B \setminus A) \cup (A \cap B)$, a disjoint union: each $x \in B$ either lies in A (so $x \in A \cap B$) or not (so $x \in B \setminus A$), and not both. By Proposition 6.33,

$$|B| = |B \setminus A| + |A \cap B|, \quad \text{so} \quad |B \setminus A| = |B| - |A \cap B|. \qquad (2)$$

Step 3. Substituting (2) into (1):

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

This is Corollary 6.34 from lectures.

> **Check:** 30 students: 18 take option X, 15 take option Y, and 7 take both. How many take neither?
> **Answer:** $30 - (18 + 15 - 7) = 30 - 26 = 4$.

MAIN ROUTE · 35–41 elapsed · 6 min

P7 · A bijection from (0,1) onto the whole real line

Pure PS6 B4

B4. Recall that in Question B2 on Problem Sheet 5, we showed that the function

$$g\colon \mathbb{R} \to (-1,1), \qquad g(x) = \frac{x}{1+|x|}$$

is bijective. Using this, prove that there exists a bijection $\varphi\colon (0,1) \to \mathbb{R}$.

[Hint. First construct a bijection $f\colon (0,1) \to (-1,1)$.] (Here $(a,b) = \{x \in \mathbb{R} : a < x < b\}$.)

> **Say:** We already have a bijection between $\mathbb{R}$ and $(-1,1)$, so the only new job is to stretch $(0,1)$ onto $(-1,1)$ with a straight line, then go back to $\mathbb{R}$ with g^{-1}. Watch the order: $\varphi = g^{-1} \circ f$ means do f first.

> 板书顶端先画箭头图 (0,1) —f→ (−1,1) —g⁻¹→ R，之后所有复合都对着这张图看方向。常见错误：写成 g∘f（落在 (−1,1) 而不是 R）或 f∘g⁻¹（定义域对不上）；以及只写 f(x) = 2x − 1，不验证它真的映到 (−1,1)、不证双射。追问：φ(3/4) 等于多少？（1）时间紧：f 的双射性用有逆函数 k 一句带过，g⁻¹ 的显式公式只口头核对。

Students write a first step (30–60 s) → then board.

Plan. $(0,1) \xrightarrow{f} (-1,1) \xrightarrow{g^{-1}} \mathbb{R}$, and $\varphi = g^{-1} \circ f$ (do f first).

Step 1: a bijection $f\colon (0,1) \to (-1,1)$. Define $f(x) = 2x - 1$. If $0 < x < 1$ then $0 < 2x < 2$, so $-1 < 2x - 1 < 1$; thus f does map $(0,1)$ into $(-1,1)$.

Define $k\colon (-1,1) \to (0,1)$ by $k(y) = \dfrac{y+1}{2}$. If $-1 < y < 1$ then $0 < y + 1 < 2$, so $0 < \frac{y+1}{2} < 1$.

$$k(f(x)) = \frac{(2x-1)+1}{2} = x \ \ (x \in (0,1)), \qquad f(k(y)) = 2 \cdot \frac{y+1}{2} - 1 = y \ \ (y \in (-1,1)).$$

So $k \circ f = \mathrm{id}_{(0,1)}$ and $f \circ k = \mathrm{id}_{(-1,1)}$: f has an inverse, hence f is a bijection.

Step 2. Since $g\colon \mathbb{R} \to (-1,1)$ is a bijection, it has an inverse $g^{-1}\colon (-1,1) \to \mathbb{R}$, which is also a bijection.

Step 3. Let $\varphi = g^{-1} \circ f\colon (0,1) \to \mathbb{R}$. This makes sense because f lands in $(-1,1)$, exactly where g^{-1} starts. A composition of bijections is a bijection (Corollary 6.19), so φ is a bijection and $|(0,1)| = |\mathbb{R}|$.

Check with an explicit formula. $g^{-1}(y) = \dfrac{y}{1-|y|}$: for $y \in (-1,1)$ put $x = \frac{y}{1-|y|}$. Then $|x| = \frac{|y|}{1-|y|}$, so $1 + |x| = \frac{1}{1-|y|}$ and $g(x) = \frac{x}{1+|x|} = y$. Hence

$$\varphi(x) = \frac{2x-1}{1-|2x-1|} = \begin{cases} \dfrac{2x-1}{2x}, & 0 < x \leqslant \frac{1}{2}, \\ \dfrac{2x-1}{2-2x}, & \frac{1}{2} \leqslant x < 1. \end{cases}$$

For example $\varphi(\frac{1}{4}) = -1$, $\varphi(\frac{1}{2}) = 0$, $\varphi(\frac{3}{4}) = 1$; $\varphi(x) \to -\infty$ as $x \to 0^+$ and $\varphi(x) \to +\infty$ as $x \to 1^-$.

> **Check:** Compute $\varphi(3/4)$, using $g^{-1}(y) = \dfrac{y}{1-|y|}$.
> **Answer:** $f(3/4) = 1/2$ and $g^{-1}(1/2) = \frac{1/2}{1/2} = 1$, so $\varphi(3/4) = 1$. (Check: $g(1) = 1/2$.)

MAIN ROUTE · 41–48 elapsed · 7 min

P8 · Composing permutations of {1,2,3}: order matters

Lecture 7.1.1 (3 Nov) + Pure PS6 B5 (partial)

In lectures the six permutations (bijections $X \to X$) of $X = \{1, 2, 3\}$ were labelled:

- e: $e(1) = 1,\ e(2) = 2,\ e(3) = 3$ (the identity)
- f: $f(1) = 1,\ f(2) = 3,\ f(3) = 2$
- g: $g(1) = 2,\ g(2) = 1,\ g(3) = 3$
- h: $h(1) = 3,\ h(2) = 1,\ h(3) = 2$
- i: $i(1) = 2,\ i(2) = 3,\ i(3) = 1$
- j: $j(1) = 3,\ j(2) = 2,\ j(3) = 1$

We write xy for $x \circ y$, so $(xy)(k) = x(y(k))$: **do y first, then x** (right to left).

(a) Compute gf and fg, and identify each as one of e, f, g, h, i, j. Is $gf = fg$?

(b) Compute $f^2 = ff$ and $i^2 = ii$.

(c) Find the inverse permutation i^{-1}.

(d) In the table whose entry in row x and column y is xy, fill in the row of f: $fe,\ ff,\ fg,\ fh,\ fi,\ fj$.

> **Say:** Composition reads right to left: fg means do g first, then f. Follow each of 1, 2, 3 through both maps, then match the result against the list of six.

> 板书顶端固定写六个置换的取值（严格按讲义标号）和 (xy)(k) = x(y(k))：先 y 后 x。先示范 gf = i（讲义原例），学生自己算 fg = h。注意两处讲义易混点：幻灯片第 32 页动机例子里的 g（1→2→3→1）是循环，与第 34 页正式标号的 g 不同（正式表里那个循环叫 i）；幻灯片箭头图的下排是重排后的位置顺序，不是 h(1),h(2),h(3)（h 的图下排是 2 3 1，但 h(1)=3,h(2)=1,h(3)=2），按函数值算。常见错误：从左往右复合（把 fg 算成 gf）。追问：哪些 x 满足 x² = e？时间紧：(d) 只做 fh、fi 两格。

Students write a first step (30–60 s) → then board.

(a) gf: $(gf)(1) = g(f(1)) = g(1) = 2$, $(gf)(2) = g(f(2)) = g(3) = 3$, $(gf)(3) = g(f(3)) = g(2) = 1$. So gf sends $1 \mapsto 2,\ 2 \mapsto 3,\ 3 \mapsto 1$: $gf = i$.

fg: $(fg)(1) = f(g(1)) = f(2) = 3$, $(fg)(2) = f(g(2)) = f(1) = 1$, $(fg)(3) = f(g(3)) = f(3) = 2$. So $fg = h$.

Since $i(1) = 2 \neq 3 = h(1)$, we have $gf \neq fg$: the order of composition matters.

(b) ff: $1 \mapsto f(1) = 1$, $2 \mapsto f(3) = 2$, $3 \mapsto f(2) = 3$. So $f^2 = e$ (swapping twice does nothing).

ii: $1 \mapsto i(2) = 3$, $2 \mapsto i(3) = 1$, $3 \mapsto i(1) = 2$. So $i^2 = h$.

(c) i sends $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 1$. Reversing the arrows, i^{-1} sends $2 \mapsto 1$, $3 \mapsto 2$, $1 \mapsto 3$, i.e. $i^{-1}(1) = 3,\ i^{-1}(2) = 1,\ i^{-1}(3) = 2$. So $i^{-1} = h$.

Check: $(ih)(1) = i(3) = 1$, $(ih)(2) = i(1) = 2$, $(ih)(3) = i(2) = 3$, so $ih = e$; similarly $hi = e$. (Also $i^2 = h = i^{-1}$, so $i^3 = e$.)

(d) $fe = f$, $ff = e$ by (b), $fg = h$ by (a), and

$$
\begin{aligned}
fh:&\ 1\mapsto f(3)=2,\ \ 2\mapsto f(1)=1,\ \ 3\mapsto f(2)=3, &&\text{so } fh=g,\\
fi:&\ 1\mapsto f(2)=3,\ \ 2\mapsto f(3)=2,\ \ 3\mapsto f(1)=1, &&\text{so } fi=j,\\
fj:&\ 1\mapsto f(3)=2,\ \ 2\mapsto f(2)=3,\ \ 3\mapsto f(1)=1, &&\text{so } fj=i.
\end{aligned}
$$

xy	e	f	g	h	i	j
f	f	e	h	g	j	i

Every one of the six permutations appears exactly once in this row.

> **Check:** Which of the six permutations x satisfy $xx=e$?
> **Answer:** e, f, g, j (the identity and the three swaps); $hh=i$ and $ii=h$.

MAIN ROUTE · 48–52 elapsed · 4 min

P9 · Symmetries of a square as permutations of its vertices

Pure PS6 A8

A8. Describe all the symmetries of a square, both geometrically (i.e. in terms of rotations and reflections) and in terms of adjacency preserving permutations of the set of vertices of the square.

Label the vertices as on the sheet (1 top right, 2 top left, 3 bottom left, 4 bottom right):

2	1
3	4

so the edges are $\{1,2\},\{2,3\},\{3,4\},\{4,1\}$, and $\{1,3\},\{2,4\}$ are diagonals. A permutation σ of $\{1,2,3,4\}$ is *adjacency preserving* if whenever vertices a and b are joined by an edge, $\sigma(a)$ and $\sigma(b)$ are also joined by an edge.

(Follow-up: why is the permutation that swaps 1 and 2 and fixes 3 and 4 not a symmetry?)

> **Say:** A symmetry of the square is a rearrangement of the corners that keeps neighbours as neighbours. Choose where corner 1 goes, 4 options; then its neighbour 2 has only 2 options, and after that everything is forced.

> 板书先画正方形并按讲义标 2 1 / 3 4（1 右上，逆时针 1→2→3→4），写出 4 条边。计数论证 4 × 2 = 8 是核心，比逐个画图快。官方解答列表打成了 id, Rπ/2, Rπ, R3π/2, ρ1, ρ2, δ2, δ2，应为 δ1, δ2（δ1 交换 1、3；δ2 交换 2、4），讲时直接用正确版。常见错误：两行记号下排写成转完后各位置上的标号（那是逆置换）；强调下排是 σ(k)，即顶点 k 去哪里。时间紧：只写 R_{π/2}、ρ1、δ1 三个的两行记号，其余口头列出。

Students write a first step (30–60 s) → then board.

How many are there? Let σ be adjacency preserving. $\sigma(1)$ can be any of the 4 vertices. Vertex 2 is joined to 1, so $\sigma(2)$ must be one of the 2 neighbours of $\sigma(1)$. Vertex 4 is also joined to 1, and $\sigma(4) \neq \sigma(2)$ because σ is injective, so $\sigma(4)$ is the other neighbour of $\sigma(1)$. Then $\sigma(3)$ is the remaining vertex. So there are at most $4 \cdot 2 = 8$ adjacency preserving permutations. All 8 below occur, since rotations and reflections of the square send edges to edges. So there are exactly 8.

The list (two-row notation: the bottom entry under k is $\sigma(k)$, the vertex to which k is moved; with this labelling $1,2,3,4$ go round anticlockwise):

name	geometric description	permutation of vertices
id	do nothing	$\begin{pmatrix}1&2&3&4\\1&2&3&4\end{pmatrix}$
$R_{\pi/2}$	rotate anticlockwise by $\pi/2$	$\begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix}$
R_{π}	rotate by π	$\begin{pmatrix}1&2&3&4\\3&4&1&2\end{pmatrix}$

$R_{3\pi/2}$	rotate anticlockwise by $3\pi/2$	$\begin{pmatrix}1&2&3&4\\4&1&2&3\end{pmatrix}$
ρ_1	reflect in the vertical line through the centre	$\begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix}$
ρ_2	reflect in the horizontal line through the centre	$\begin{pmatrix}1&2&3&4\\4&3&2&1\end{pmatrix}$
δ_1	reflect in the diagonal through 2 and 4	$\begin{pmatrix}1&2&3&4\\3&2&1&4\end{pmatrix}$
δ_2	reflect in the diagonal through 1 and 3	$\begin{pmatrix}1&2&3&4\\1&4&3&2\end{pmatrix}$

So the eight symmetries are $\mathrm{id}, R_{\pi/2}, R_{\pi}, R_{3\pi/2}, \rho_1, \rho_2, \delta_1, \delta_2$: four rotations and four reflections.

Follow-up. Let $\tau = \begin{pmatrix}1&2&3&4\\2&1&3&4\end{pmatrix}$. Then the edge $\{2,3\}$ is sent to $\{\tau(2),\tau(3)\} = \{1,3\}$, which is a diagonal, not an edge. So τ is not adjacency preserving: you cannot move the square rigidly so that two neighbouring corners swap while the other two stay put. It is still a permutation of $\{1,2,3,4\}$ (there are $4! = 24$ of those), but only 8 of the 24 preserve the square.

> **Check:** Is $\begin{pmatrix}1&2&3&4\\1&3&2&4\end{pmatrix}$ a symmetry of the square?
> **Answer:** No: the edge $\{1,2\}$ is sent to $\{1,3\}$, which is a diagonal.

52–55 · EXIT CHECK

Three quick checks before you leave

1. Can a function from a 10-element set to a 9-element set be injective? Name the principle.

2. Give a bijection $\mathbb{Z} \to 2\mathbb{Z}$. What does it show about proper subsets of infinite sets?

3. With the lecture labels, is $fg = gf$?

1. No — Pigeonhole Principle (more inputs than possible outputs forces a collision).

2. $n \mapsto 2n$ (inverse $m \mapsto m/2$). An infinite set can have the same cardinality as a proper subset; this never happens for finite sets.

3. No: $gf = i$ but $fg = h$.

> 最后三分钟口答即可；第 3 题让学生说出“先做右边”。离开前再提醒一次 PS5 peer feedback 13:00 截止，以及 PS6 Assigned Question 下周四 13:00 交。

Reserve answer bank — Pure

For questions from the room or early finishers. Hand-in questions show hints only.

RESERVE / LOOKUP

R-A3 · Reserve: size of a Cartesian product

Pure PS6 A3

A3. Let A and B be non-empty finite sets. Prove that $|A \times B| = |A||B|$, where $A \times B$ is the Cartesian product of A and B.

> **Say:** Arrange the pairs in a grid with m rows and n columns. Counting the grid row by row gives a bijection onto $\{1, \ldots, mn\}$.

> 备用题（Lecture 1 建议题）。直觉版（m 行 n 列网格）口头讲，严谨版写 F(a_i,b_j) = n(i−1)+j 并证单射、满射（满射用带余除法）。学生常只写 A×B 显然有 mn 个元素；追问：为什么不同的 (i,j) 给出不同的有序对？（有序对相等当且仅当两个分量都相等）

Let $|A| = m$, $|B| = n$, and list $A = \{a_1, \ldots, a_m\}$, $B = \{b_1, \ldots, b_n\}$ without repetition. Recall $A \times B = \{(a, b) : a \in A,\ b \in B\}$ and $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$. Define

$$F\colon A \times B \to \{1, 2, \ldots, mn\}, \qquad F((a_i, b_j)) = n(i-1) + j.$$

Well defined: for $1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$ we get $1 \leqslant n(i-1) + j \leqslant n(m-1) + n = mn$.

Injective: if $n(i-1) + j = n(i'-1) + j'$, then $j - j' = n(i'-i)$ is a multiple of n; but $|j - j'| \leqslant n - 1$, so $j = j'$, and then $i = i'$. Hence $(a_i, b_j) = (a_{i'}, b_{j'})$.

Surjective: let $k \in \{1, \ldots, mn\}$. By the division algorithm, $k - 1 = nq + r$ with $q, r \in \mathbb{Z}$, $0 \leqslant r \leqslant n - 1$. From $nq \leqslant k - 1 \leqslant mn - 1$ we get $q \leqslant m - 1$, and from $nq = k - 1 - r \geqslant -(n-1)$ we get $q \geqslant 0$. Put $i = q + 1 \in \{1, \ldots, m\}$, $j = r + 1 \in \{1, \ldots, n\}$; then $F((a_i, b_j)) = nq + r + 1 = k$.

So F is a bijection and $|A \times B| = mn = |A||B|$.

(Alternative: $A \times B$ is the disjoint union of the m sets $\{a_i\} \times B$, each of size n, so B2 gives $|A \times B| = n + \cdots + n = mn$.)

> **Check:** If $|A| = 3$ and $|B| = 4$, what is $|A \times B \times A|$?
> **Answer:** $3 \cdot 4 \cdot 3 = 36$.

RESERVE / LOOKUP

R-B2 · Reserve: union of n pairwise disjoint sets (induction)

Pure PS6 B2

B2. Let $n \geqslant 2$ be an integer and let $A_1, \ldots, A_n$ be non-empty finite sets that are pairwise disjoint, so $A_i \cap A_j = \emptyset$ for all $i, j \in \{1, \ldots, n\}$ with $i \neq j$. Use induction on n to prove that

$$|A_1 \cup \cdots \cup A_n| = |A_1| + \cdots + |A_n|.$$

[You may use Proposition 6.33: if A and B are disjoint finite subsets of a set X, then $|A \cup B| = |A| + |B|$.]

> **Say:** Glue the first k sets into one big set U. Then you only ever need the two-set result, Proposition 6.33.

> 备用题（Lecture 2 建议题）。板书结构：P(n) 的完整陈述 → base case n = 2 即 Prop 6.33 → 归纳步。学生最常漏的是证明 U 与 A_{k+1} 不相交（要用 pairwise disjoint）。追问：只要求 A1∩A2∩⋯∩An = ∅ 行不行？（不行：A1 = A2 = {1}, A3 = {2}，三者交集为空，但并集大小 2 ≠ 3）

For each integer $n \geqslant 2$ let $P(n)$ be the statement: for any non-empty, finite, pairwise disjoint sets $A_1, \ldots, A_n$, $|A_1 \cup \cdots \cup A_n| = |A_1| + \cdots + |A_n|$.

Base case $n = 2$**.** This is Proposition 6.33 (with $X = A_1 \cup A_2$).

Inductive step. Let $k \geqslant 2$ and assume $P(k)$. Let $A_1, \ldots, A_{k+1}$ be non-empty, finite and pairwise disjoint, and put $U = A_1 \cup \cdots \cup A_k$. The sets $A_1, \ldots, A_k$ are pairwise disjoint, so by $P(k)$, U is finite with $|U| = |A_1| + \cdots + |A_k|$.

U and A_{k+1} are disjoint: if $x \in U \cap A_{k+1}$, then $x \in A_i$ for some $i \leqslant k$, so $x \in A_i \cap A_{k+1} = \emptyset$, which is impossible. Hence, by Proposition 6.33,

$$|A_1 \cup \cdots \cup A_{k+1}| = |U \cup A_{k+1}| = |U| + |A_{k+1}| = |A_1| + \cdots + |A_k| + |A_{k+1}|,$$

which is $P(k+1)$. By the Principle of Mathematical Induction, $P(n)$ holds for every integer $n \geqslant 2$.

> **Check:** In the inductive step, which hypothesis guarantees $U \cap A_{k+1} = \emptyset$?
> **Answer:** Pairwise disjointness: $A_i \cap A_{k+1} = \emptyset$ for every $i \leqslant k$.

RESERVE / LOOKUP

R-C4 · Reserve: pigeonhole + Archimedes (Dirichlet-type approximation)

Pure PS6 C4

C4. Let $x \in \mathbb{R}$ be an irrational number. Show that for any $\varepsilon > 0$, there exist $m, n \in \mathbb{Z}$ with $m \neq 0$ so that $|mx + n| < \varepsilon$.

[Hint. Choose $N \in \mathbb{Z}_+$ with $1/N < \varepsilon$. Recall that if $y \in \mathbb{R}$ is a real number, then $\lfloor y \rfloor \in \mathbb{Z}$ is the unique integer such that $\lfloor y \rfloor \leqslant y < \lfloor y \rfloor + 1$. Consider the real numbers

$$0x - \lfloor 0x \rfloor,\ 1x - \lfloor 1x \rfloor,\ \ldots,\ Nx - \lfloor Nx \rfloor$$

and the intervals $[0, 1/N),\ [1/N, 2/N),\ \ldots,\ [(N-1)/N, 1)$, and use the pigeonhole principle.]

> **Say:** Look only at the fractional parts of $0x, 1x, \ldots, Nx$: that is $N + 1$ numbers in $[0, 1)$, and we chop $[0, 1)$ into N little boxes. Two of them must land in the same box.

> C 题（Lecture 2 建议题），给做得快的学生或课后。板书：先画 [0,1) 切成 N 段，标 N+1 个点 t_0,…,t_N：N+1 只鸽子，N 个笼子。常见错误：用闭区间导致区间重叠；忘了 m ≠ 0 来自 i ≠ j。追问：证明里哪里用到了 x 是无理数？（没用到；有理数时可直接取到 0）官方解答小笔误：the N + 1 reals numbers 应为 real numbers。

Let $\varepsilon > 0$. By the Archimedean Property choose $N \in \mathbb{Z}_+$ with $N > 1/\varepsilon$, i.e. $1/N < \varepsilon$.

For $k \in \{0, 1, \ldots, N\}$ put $t_k = kx - \lfloor kx \rfloor$. Since $\lfloor kx \rfloor \leqslant kx < \lfloor kx \rfloor + 1$, we have $0 \leqslant t_k < 1$.

The intervals $I_r = [r/N, (r+1)/N)$, $r = 0, 1, \ldots, N-1$, are pairwise disjoint and their union is $[0, 1)$. So each t_k lies in exactly one I_r; let $p(k)$ be that r. This defines $p\colon \{0, 1, \ldots, N\} \to \{0, 1, \ldots, N-1\}$.

The domain has $N + 1$ elements and the codomain has N, so by the Pigeonhole Principle p is not injective: there are $i \neq j$ in $\{0, \ldots, N\}$ such that t_i and t_j lie in the same interval I_r. Two points of a half-open interval of length $1/N$ are less than $1/N$ apart, so $|t_i - t_j| < 1/N$.

Now $t_i - t_j = (i - j)x + (\lfloor jx \rfloor - \lfloor ix \rfloor)$. Put $m = i - j$ and $n = \lfloor jx \rfloor - \lfloor ix \rfloor$. Both are integers, and $m \neq 0$ because $i \neq j$. Then

$$|mx + n| = |t_i - t_j| < \frac{1}{N} < \varepsilon.$$

Remark. Irrationality was not used: for rational $x = p/q$ one can even take $m = q$, $n = -p$ to get 0. The interesting point for irrational x is that $mx + n$ is never 0 when $m \neq 0$, yet it can be made as small as we like.

> **Check:** Where does the condition $m \neq 0$ come from in the proof?
> **Answer:** From $i \neq j$: pigeonhole gives two different indices, and $m = i - j$.

RESERVE / LOOKUP

R-A9 · Reserve (preview): is it a binary operation?

Pure PS6 A9

A9. Let $R = \{a + bi : a, b \in \mathbb{Z}_+\} \subseteq \mathbb{C}$ and let $+$ and $\times$ denote the usual addition and multiplication operations on the set $\mathbb{C}$ of complex numbers. Is $+$ a binary operation on R? What about $\times$?

[Preview: a binary operation on a set S is a function $S \times S \to S$; so the question is whether the sum, or product, of two elements of R always lies in R.]

> **Say:** A binary operation on a set must keep you inside the set. So test it: is the sum of two elements of R still in R? Is the product?

> 预告题（二元运算在 Chapter 7 后面才正式定义），给提前做完的学生。关键：说不是需要一个具体反例，(1+i)² = 2i，实部 0 不是正整数。常见错误：忘了 Z+ 不含 0。追问：把 Z+ 换成 Z，乘法是否封闭？（是）

$+$: yes. Let $a + bi,\ c + di \in R$, so $a, b, c, d \in \mathbb{Z}_+$. Then $(a + bi) + (c + di) = (a + c) + (b + d)i$, and $a + c,\ b + d \in \mathbb{Z}_+$ (sums of positive integers are positive integers). So the sum lies in R, and $+$ is a binary operation on R.

$\times$: no. $1 + i \in R$, but

$$(1 + i) \times (1 + i) = 1 + 2i + i^2 = 0 + 2i,$$

whose real part 0 is not in $\mathbb{Z}_+$. So $(1 + i) \times (1 + i) \notin R$, and $\times$ is not a binary operation on R. (In general $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ can be $\leqslant 0$.)

> **Check:** If $\mathbb{Z}_+$ is replaced by $\mathbb{Z}$, is $\times$ a binary operation on the new set?
> **Answer:** Yes: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ with $ac - bd,\ ad + bc \in \mathbb{Z}$.

RESERVE / LOOKUP

R-B5 · Reserve: full Cayley table of the permutations of {1,2,3}

Pure PS6 B5

B5. Let $X = \{1, 2, 3\}$ and let G be the group of permutations of X. Choose an ordering of the elements in G and construct the corresponding Cayley table of G.

(Use the lecture labels below. The Cayley table has the entry $xy = x \circ y$, meaning do y first, in row x and column y. The word "group" will be defined in a coming lecture; here G is the set of the six permutations with composition.)

- e: $e(1) = 1,\ e(2) = 2,\ e(3) = 3$ (the identity)
- f: $f(1) = 1,\ f(2) = 3,\ f(3) = 2$
- g: $g(1) = 2,\ g(2) = 1,\ g(3) = 3$
- h: $h(1) = 3,\ h(2) = 1,\ h(3) = 2$
- i: $i(1) = 2,\ i(2) = 3,\ i(3) = 1$
- j: $j(1) = 3,\ j(2) = 2,\ j(3) = 1$

> **Say:** Fill the table one row at a time: row x, column y is xy, so do y first. A good check: every row and every column must contain each permutation exactly once.

> 备用（P8 的延伸）。按讲义标号 e, f, g, h, i, j 排序；黑板上写清约定：行 x 列 y 填 xy = x∘y（先 y 后 x）。整张表已逐格独立验算，与官方解答一致。提醒：讲义箭头图的下排不是函数值（h 的图下排是 2 3 1，但 h(1)=3, h(2)=1, h(3)=2）。快速自检：每行每列都不重复；两个对换之积是 e, h, i 中之一。

Ordering e, f, g, h, i, j. Entry in row x, column y is $xy = x \circ y$ (row = left factor = applied last).

$\circ$	e	f	g	h	i	j
e	e	f	g	h	i	j
f	f	e	h	g	j	i
g	g	i	e	j	f	h
h	h	j	f	i	e	g
i	i	g	j	e	h	f
j	j	h	i	f	g	e

Sample entries (each entry is found by following 1, 2, 3):

- fg: $1 \mapsto f(2) = 3,\ 2 \mapsto f(1) = 1,\ 3 \mapsto f(3) = 2$, so $fg = h$; and $gf = i$ (lecture).
- gh: $1 \mapsto g(3) = 3,\ 2 \mapsto g(1) = 2,\ 3 \mapsto g(2) = 1$, so $gh = j$.
- hg: $1 \mapsto h(2) = 1,\ 2 \mapsto h(1) = 3,\ 3 \mapsto h(3) = 2$, so $hg = f$.
- hh: $1 \mapsto h(3) = 2,\ 2 \mapsto h(1) = 3,\ 3 \mapsto h(2) = 1$, so $hh = i$; and $hi = ih = e$.
- ij: $1 \mapsto i(3) = 1,\ 2 \mapsto i(2) = 3,\ 3 \mapsto i(1) = 2$, so $ij = f$; ji: $1 \mapsto j(2) = 2,\ 2 \mapsto j(3) = 1,\ 3 \mapsto j(1) = 3$, so $ji = g$.

Checks and observations.

- Every row and every column contains each of the six permutations exactly once.
- The table is not symmetric about the diagonal: $fg = h \neq i = gf$.
- Diagonal: $ff = gg = jj = e$ (the three swaps undo themselves), $hh = i$, $ii = h$; so $h^{-1} = i$ and $i^{-1} = h$.
- Structure: e, h, i are the rotations of three points in a circle, f, g, j the swaps. Two swaps give a rotation; a swap and a rotation give a swap.

> **Check:** From the table, what is jg, and is it equal to gj?
> **Answer:** $jg = i$ but $gj = h$, so no.

RESERVE / LOOKUP

R-B7 · Reserve: symmetries of a rectangle and their Cayley table

Pure PS6 B7

B7. Let R be the following rectangle (wider than it is tall):

2	(long edge)	1
(short edge)		(short edge)
3	(long edge)	4

so $\{1,2\}$ and $\{3,4\}$ are the long edges and $\{2,3\}$ and $\{1,4\}$ are the short edges. Define a *symmetry* of R to be a distance-preserving permutation f of the set of vertices $\{1,2,3,4\}$: this means that if vertices i and j are joined by a short edge, then their images $f(i)$ and $f(j)$ are also joined by a short edge, and similarly for long edges.

List all the symmetries of R. And for every pair of symmetries f and g, calculate $f \circ g$, presenting your results in the form of a Cayley table.

> **Say:** First count: corner 1 can go anywhere, but then its long-edge neighbour and its short-edge neighbour are forced. So there are at most four symmetries. Find them, then compose.

备用题，与 P9 正方形对比：正方形 8 个，长方形只有 4 个（旋转 π/2 和对角线反射会把长边变成短边）。表的约定同 B5：行 f 列 g 填 f∘g（先 g）。值得点出：这张表关于对角线对称（可交换）且对角线全是 id，和 {1,2,3} 的置换表不同。

Counting. $f(1)$ can be any of the 4 vertices. Vertex 2 is joined to 1 by a long edge, and each vertex has exactly one long-edge neighbour, so $f(2)$ is forced; likewise $f(4)$ is the unique short-edge neighbour of $f(1)$, and $f(3)$ is the remaining vertex. So there are at most 4 symmetries, and the following 4 all work:

- $\mathrm{id} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$ (do nothing);
- $R_\pi = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ (rotate by π about the centre);
- $\rho_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$ (reflect in the vertical line through the centre);
- $\rho_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$ (reflect in the horizontal line through the centre).

(Rotation by $\pi/2$ is not a symmetry: it sends the long edge $\{1,2\}$ to the short edge $\{2,3\}$. The diagonal reflections fail similarly.)

Cayley table (row f, column g, entry $f \circ g$: do g first):

$\circ$	id	R_π	ρ_1	ρ_2
id	id	R_π	ρ_1	ρ_2
R_π	R_π	id	ρ_2	ρ_1
ρ_1	ρ_1	ρ_2	id	R_π
ρ_2	ρ_2	ρ_1	R_π	id

ρ_2	ρ_2	ρ_1	R_π	id

Example: $R_\pi \circ \rho_1$: $1 \mapsto 2 \mapsto 4$, $2 \mapsto 1 \mapsto 3$, $3 \mapsto 4 \mapsto 2$, $4 \mapsto 3 \mapsto 1$, giving $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} = \rho_2$. Similarly $\rho_1 \circ \rho_2$: $1 \mapsto 4 \mapsto 3$, $2 \mapsto 3 \mapsto 4$, $3 \mapsto 2 \mapsto 1$, $4 \mapsto 1 \mapsto 2$, giving R_π.

Observations: every symmetry composed with itself is id, and the table is symmetric about the diagonal, so here $f \circ g = g \circ f$ for all pairs, unlike the permutations of $\{1, 2, 3\}$.

> **Check:** Why is rotation by $\pi/2$ not a symmetry of the rectangle?
> **Answer:** It sends the long edge $\{1, 2\}$ to $\{2, 3\}$, which is a short edge.

RESERVE / LOOKUP

R-B6 · Preview (groups not yet lectured): the only group with three elements

Pure PS6 B6 (preview, not hand-in)

B6. Let $G = \{e, x, y\}$ be a set with three elements. Show that there is exactly one binary operation $\star\colon G \times G \to G$ such that $(G, \star)$ is a group with identity element e.

Preview (groups have not been defined in lectures yet). A **group** $(G, \star)$ is a set G with a binary operation $\star\colon G \times G \to G$ such that: (G1) $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$; (G2) there is an identity $e \in G$ with $e \star a = a \star e = a$ for all $a \in G$; (G3) every $a \in G$ has an inverse $a^{-1} \in G$ with $a \star a^{-1} = a^{-1} \star a = e$. The group is **abelian** if $a \star b = b \star a$ for all $a, b \in G$.

> **Say:** Preview only: groups are not defined yet, so treat this as a puzzle. The key tool is cancellation: in a group $ab = ac$ forces $b = c$, so no row of the table can repeat an element.

> 预告题，群的定义尚未讲，只在学生主动问时用，不是交作业题。思路：先用单位元填满第一行第一列，再用消去律逐格排除。存在性别忘：要说明结合律成立；最省事是对应到旋转 {e, i, h} 的复合表，直接用函数复合的结合律（Prop 6.17）。

Write ab for $a \star b$.

Cancellation. If $ab = ac$, then $b = eb = (a^{-1}a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = c$. Similarly $ba = ca$ implies $b = c$.

Every entry is forced.

- Since e is the identity: $ee = e$, $ex = xe = x$, $ey = ye = y$.
- xy: if $xy = x = xe$, cancelling x gives $y = e$, false; if $xy = y = ey$, cancelling y on the right gives $x = e$, false. So $xy = e$. In the same way $yx = e$.
- xx: if $xx = x = xe$ then $x = e$, false; if $xx = e = xy$ then $x = y$, false. So $xx = y$.
- yy: if $yy = y = ye$ then $y = e$, false; if $yy = e = yx$ then $y = x$, false. So $yy = x$.

$\star$	e	x	y
e	e	x	y
x	x	y	e
y	y	e	x

So there is *at most one* such operation.

This table really is a group. e is an identity, and $e^{-1} = e$, $x^{-1} = y$, $y^{-1} = x$. For associativity, compare with the permutations e, i, h of $\{1, 2, 3\}$: from the lecture labels, $ii = h$, $hh = i$, $ih = hi = e$. So under $e \mapsto e$, $x \mapsto i$, $y \mapsto h$ the table above is exactly the composition table of $\{e, i, h\}$, and composition of functions is associative (Proposition 6.17). Hence $\star$ is associative. (Alternatively, $e \mapsto 0$, $x \mapsto 1$, $y \mapsto 2$ turns $\star$ into addition modulo 3.)

Therefore there is exactly one such binary operation.

Check: In a group, can one row of the Cayley table contain the same element twice?
Answer: No: if $ab = ac$, multiplying on the left by a^{-1} gives $b = c$.

RESERVE / LOOKUP

R-B8 · Preview (groups not yet lectured): a group of upper-triangular matrices

Pure PS6 B8 (preview, not hand-in)

B8. Let G be the following set of 2×2 matrices with rational entries:

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & \frac{1}{a} \end{pmatrix} : a \in \mathbb{Q} \setminus \{0\},\ b \in \mathbb{Q} \right\}.$$

Prove that G is a group with respect to matrix multiplication. Is G abelian?

Preview (groups have not been defined in lectures yet). A **group** $(G, \star)$ is a set G with a binary operation $\star\colon G \times G \to G$ such that: (G1) $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$; (G2) there is an identity $e \in G$ with $e \star a = a \star e = a$ for all $a \in G$; (G3) every $a \in G$ has an inverse $a^{-1} \in G$ with $a \star a^{-1} = a^{-1} \star a = e$. The group is **abelian** if $a \star b = b \star a$ for all $a, b \in G$.

> **Say:** Preview: check four things: the product stays in G, multiplication is associative, the identity matrix is in G, and each matrix has an inverse of the same shape.

> 预告题，与线性代数课的矩阵乘法衔接，不是交作业题。最容易漏的是：封闭性里右下角必须恰好是 1/(ac)；逆矩阵也要验证属于 G（参数 1/a ≠ 0）。非交换用最简单的一对 diag(2, 1/2) 和 (1 1; 0 1)。

Closure. For $a, c \in \mathbb{Q} \setminus \{0\}$ and $b, d \in \mathbb{Q}$,

$$\begin{pmatrix} a & b \\ 0 & \frac{1}{a} \end{pmatrix} \begin{pmatrix} c & d \\ 0 & \frac{1}{c} \end{pmatrix} = \begin{pmatrix} ac & ad + \frac{b}{c} \\ 0 & \frac{1}{ac} \end{pmatrix}.$$

Here $ac \in \mathbb{Q} \setminus \{0\}$, $ad + \frac{b}{c} \in \mathbb{Q}$, and the bottom-right entry is exactly $\frac{1}{ac}$; so the product lies in G.

Associativity. Matrix multiplication is associative.

Identity. Taking $a = 1, b = 0$ gives $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in G$, and $IM = MI = M$ for all M.

Inverses. For $M = \begin{pmatrix} a & b \\ 0 & \frac{1}{a} \end{pmatrix}$ let $M' = \begin{pmatrix} \frac{1}{a} & -b \\ 0 & a \end{pmatrix}$. $M' \in G$: its parameters are $\frac{1}{a} \in \mathbb{Q} \setminus \{0\}$ and $-b \in \mathbb{Q}$, and its bottom-right entry is $a = 1/\frac{1}{a}$. Moreover

$$MM' = \begin{pmatrix} a \cdot \frac{1}{a} & -ab + ba \\ 0 & \frac{1}{a} \cdot a \end{pmatrix} = I, \qquad M'M = \begin{pmatrix} \frac{1}{a} \cdot a & \frac{b}{a} - \frac{b}{a} \\ 0 & a \cdot \frac{1}{a} \end{pmatrix} = I.$$

Hence G is a group.

Not abelian. Let $A = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, both in G. Then

$$AB = \begin{pmatrix} 2 & 2 \\ 0 & \frac{1}{2} \end{pmatrix}, \qquad BA = \begin{pmatrix} 2 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix},$$

so $AB \neq BA$ and G is not abelian. (The official example $\begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 6 & \frac{7}{3} \\ 0 & \frac{1}{6} \end{pmatrix} \neq \begin{pmatrix} 6 & \frac{7}{2} \\ 0 & \frac{1}{6} \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$ also works.)

Check: What is the inverse of $\begin{pmatrix} 2 & 3 \\ 0 & \frac{1}{2} \end{pmatrix}$ in G?

Answer: $\begin{pmatrix} \frac{1}{2} & -3 \\ 0 & 2 \end{pmatrix}$.

RESERVE / LOOKUP

R-C1 · Preview (groups not yet lectured): brackets do not matter

Pure PS6 C1 (preview, not hand-in)

C1. Let G be a multiplicatively-written group. For each $n \in \mathbb{Z}_+$, consider the following statement:

$P(n)$: For all $a_1, \ldots, a_n \in G$, the product $a_1 a_2 \cdots a_n \in G$ does not depend on the positioning of parentheses.

For example, $P(4)$ asserts that for all elements $a, b, c, d \in G$, we have

$$(a(bc))d = ((ab)c)d = (ab)(cd) = a((bc)d) = a(b(cd)).$$

Use induction and associativity in G to prove that $P(n)$ is true for all $n \in \mathbb{Z}_+$.

Preview (groups have not been defined in lectures yet). A **group** $(G, \star)$ is a set G with a binary operation $\star\colon G \times G \to G$ such that: (G1) $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$; (G2) there is an identity $e \in G$ with $e \star a = a \star e = a$ for all $a \in G$; (G3) every $a \in G$ has an inverse $a^{-1} \in G$ with $a \star a^{-1} = a^{-1} \star a = e$. The group is **abelian** if $a \star b = b \star a$ for all $a, b \in G$.

> **Say:** Preview: show that every bracketing equals the all-brackets-to-the-right product. Look at the last multiplication you do: it splits the product into a left block and a right block, both shorter.

C 题预告，群尚未讲，只给想挑战的学生，不是交作业题。题面原文 does not depend of 已改为 depend on。官方解答两处小问题：Principal of Mathematical Induction 应为 Principle；归纳假设里的 k 与分割位置 k 重名，这里改用 m 和 Q(n)。关键步骤：k ⩾ 2 时用一次三元结合律把 a_1 提到最外面。

For $a_1, \ldots, a_n \in G$ define the right-nested product $r(a_1) = a_1$ and $r(a_1, \ldots, a_n) = a_1\, r(a_2, \ldots, a_n)$, i.e. $r(a_1, \ldots, a_n) = a_1(a_2(\cdots(a_{n-1}a_n)\cdots))$. We prove by strong induction on n:

$Q(n)$: every bracketing of $a_1 a_2 \cdots a_n$ (in this order) equals $r(a_1, \ldots, a_n)$.

Clearly $Q(n)$ implies $P(n)$.

Base cases. For $n = 1, 2$ there is only one bracketing. (For $n = 3$, $(a_1a_2)a_3 = a_1(a_2a_3)$ is associativity.)

Inductive step. Let $n \geqslant 3$ and assume $Q(m)$ for all $m < n$. Let x be any bracketing of $a_1 \cdots a_n$. The last multiplication performed splits it as $x = uv$, where u is a bracketing of $a_1 \cdots a_k$ and v is a bracketing of $a_{k+1} \cdots a_n$, for some $1 \leqslant k \leqslant n-1$. Since $k < n$ and $n - k < n$, the hypothesis gives $u = r(a_1, \ldots, a_k)$ and $v = r(a_{k+1}, \ldots, a_n)$.

- If $k = 1$: $x = a_1\, r(a_2, \ldots, a_n) = r(a_1, \ldots, a_n)$.
- If $k \geqslant 2$: $u = a_1 w$ with $w = r(a_2, \ldots, a_k)$, so by associativity $x = (a_1 w)v = a_1(wv)$. Now wv is a bracketing of the $n - 1$ elements $a_2 \cdots a_n$, so by $Q(n-1)$, $wv = r(a_2, \ldots, a_n)$. Hence $x = a_1\, r(a_2, \ldots, a_n) = r(a_1, \ldots, a_n)$.

So $Q(n)$ holds. By the Principle of (strong) Mathematical Induction, $Q(n)$, and hence $P(n)$, holds for all $n \in \mathbb{Z}_+$.

Example ($n = 5$): $x = ((a_1a_2)a_3)(a_4a_5)$ has $k = 3$, $u = (a_1a_2)a_3 = a_1(a_2a_3)$, $v = a_4a_5$.

> **Check:** Which axiom is used to go from $(a_1w)v$ to $a_1(wv)$?
> **Answer:** Associativity, applied to the three elements a_1, w, v.

RESERVE / LOOKUP

R-C2 · Preview (groups not yet lectured): composing symmetries of a circle

Pure PS6 C2 (preview, not hand-in)

C2. Let G be the symmetry group of a circle. You may assume that every symmetry is either a rotation around the centre of the circle, or a reflection in a line through the centre of the circle.

For every pair f, g of symmetries, calculate the composition $f \circ g$.

> **Say:** Preview: put the circle at the origin and write each symmetry as a 2 by 2 matrix. Then composing symmetries is just multiplying matrices, and the angle-addition formulas do the rest.

预告题，联系线性代数课的旋转、反射矩阵，不是交作业题。四条结论写成一张小表即可：转∘转 = 转（角相加）；两次反射 = 旋转（夹角的两倍）；转∘反射与反射∘转都是反射但轴不同（不可交换）。时间不允许，只在学生问时给结论。

Put the circle in $\mathbb{R}^2$ centred at the origin. Let R_θ be the anticlockwise rotation through θ, and S_φ the reflection in the line through the origin making angle φ with the positive x-axis. As matrices,

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad S_\varphi = \begin{pmatrix} \cos 2\varphi & \sin 2\varphi \\ \sin 2\varphi & -\cos 2\varphi \end{pmatrix}.$$

(Check: S_φ fixes $(\cos\varphi, \sin\varphi)$ and sends $(-\sin\varphi, \cos\varphi)$ to its negative.) Composition $f \circ g$ corresponds to the matrix product (matrix of f) times (matrix of g). Using the angle-addition formulas:

$$\begin{aligned}
R_\theta R_\varphi &= \begin{pmatrix} \cos(\theta+\varphi) & -\sin(\theta+\varphi) \\ \sin(\theta+\varphi) & \cos(\theta+\varphi) \end{pmatrix} = R_{\theta+\varphi}, \\
R_\theta S_\varphi &= \begin{pmatrix} \cos(\theta+2\varphi) & \sin(\theta+2\varphi) \\ \sin(\theta+2\varphi) & -\cos(\theta+2\varphi) \end{pmatrix} = S_{\varphi+\theta/2}, \\
S_\varphi R_\theta &= \begin{pmatrix} \cos(2\varphi-\theta) & \sin(2\varphi-\theta) \\ \sin(2\varphi-\theta) & -\cos(2\varphi-\theta) \end{pmatrix} = S_{\varphi-\theta/2}, \\
S_\varphi S_\theta &= \begin{pmatrix} \cos(2\varphi-2\theta) & -\sin(2\varphi-2\theta) \\ \sin(2\varphi-2\theta) & \cos(2\varphi-2\theta) \end{pmatrix} = R_{2(\varphi-\theta)}.
\end{aligned}$$

For instance, the $(1,1)$ entry of $R_\theta S_\varphi$ is $\cos\theta\cos 2\varphi - \sin\theta\sin 2\varphi = \cos(\theta+2\varphi)$, and the $(1,2)$ entry of $S_\varphi S_\theta$ is $\cos 2\varphi \sin 2\theta - \sin 2\varphi \cos 2\theta = -\sin(2\varphi - 2\theta)$; the other entries are similar.

Summary. rotation ∘ rotation = rotation (angles add); rotation ∘ reflection and reflection ∘ rotation are reflections, but in different lines in general (e.g. $\theta = \pi/2$), so G is not abelian; reflection ∘ reflection = rotation through twice the angle between the lines. Rotation angles are taken modulo 2π, reflection angles modulo π. In particular $S_\varphi S_\varphi = R_0 = \mathrm{id}$.

> **Check:** What is $S_\varphi \circ S_\varphi$?
> **Answer:** $R_0 = \mathrm{id}$: reflecting twice in the same line does nothing.

HAND-IN · HINTS ONLY — do not give the full answer

R-AQ · Assigned Question (hand-in): which of these are groups?

Pure PS6 Assigned Question (due 13:00 Thursday 12 November; feedback only, non-assessed)

Assigned Question (submit by 13:00 on Thursday 12 November).

Which of the following are groups, and why? For those that are groups, find the identity element, find the inverse of every element, and decide whether or not the group is abelian. Please explain your answers.

(a) The set of even integers under addition.

(b) The set of odd integers under addition.

(c) The set $\mathbb{R}_+$ of positive real numbers under multiplication.

(d) $(\mathbb{Z}, \star)$, where $x \star y = x + y + 1$ for all $x, y \in \mathbb{Z}$.

(e) $(\mathbb{Z}, \bullet)$, where $x \bullet y = y$ for all $x, y \in \mathbb{Z}$.

> **Say:** This is the one to hand in by 13:00 next Thursday, so I will only give you the checklist. Go through the axioms one by one for each part, and if something fails, give a concrete counterexample with actual numbers.

> 交作业题：只给检查清单，不给结论，不当场判对错。群的正式定义尚未讲，提醒学生以讲义版本为准。截止 11 月 12 日（周四）13:00，非计分，只给反馈。学生追问某小题是不是群时，只反问：单位元要两边都成立吗？逆元在不在这个集合里？

Hand-in question: hints only, no verdicts. Use the definition of a group from the lecture notes once it has been covered. For each part, test in this order (write $*$ for the operation):

1. **Closure** (is $*$ a binary operation?): for all x, y in the set, is $x * y$ again in the set?
2. **Associativity:** is $(x * y) * z = x * (y * z)$ for all x, y, z? Expand both sides.
3. **Identity:** is there one element e in the set with $e * x = x$ *and* $x * e = x$ for every x?
4. **Inverses:** for each x, is there y *in the set* with $x * y = y * x = e$?
5. If it is a group: **abelian** means $x * y = y * x$ for all x, y.

Where to look in each part

- (a) Write even integers as $2a$ with $a \in \mathbb{Z}$ and check closure; then ask whether the natural identity candidate and the natural inverse candidate are even.
- (b) Start with closure: write odd integers as $2a + 1$ and compute the sum of two of them. Also ask whether the identity candidate is in the set.
- (c) Closure under multiplication; identity candidate 1; inverse candidate $1/x$. Check that each candidate lies in $\mathbb{R}_+$.
- (d) Expand $(x \star y) \star z$ and $x \star (y \star z)$ and compare. Find an identity candidate by solving $x \star e = x$ for e, then check $e \star x = x$ too. Find an inverse candidate by solving $x \star y = e$ for y and check that $y \in \mathbb{Z}$. Finally compare $x \star y$ with $y \star x$.
- (e) Check closure and associativity by expanding both sides. Then concentrate on the identity axiom: an identity must work on *both* sides, so test $e \bullet x = x$ and $x \bullet e = x$ separately, for all x.

A complete answer contains: a clear verdict for every part; if it is a group, the identity element, the inverse of every element (a formula), and abelian or not with a reason; if it is not a group, the axiom that fails together with an explicit counterexample using actual numbers.

> **Check:** Name the four things you must check to show that something is a group.
> **Answer:** Closure (binary operation), associativity, an identity element, inverses.

Sources and scope

Pure: Problem Sheet 6 (ex6.pdf), its Part A and full solutions, and the Part II lecture summary (Week 7 lectures of 2 and 3 November), downloaded from the MATH10027 Blackboard course on 4 November 2026.

Linear: Linear Algebra Problem Sheet 6 and its solutions, and Chapter 7 lecture notes parts 1–3 with the annotated Lectures 7.1 and 7.2, downloaded from the MATH10015 Blackboard course on 4 November 2026.

Every solution has been re-derived and checked against the official solutions; obvious typographical errors in the sources are corrected. Hand-in questions (Pure PS6 Assigned Question, due 13:00 Thursday 12 November; Linear PS6 Q1, due noon Wednesday 11 November) are hints only. The student worksheets contain questions and the key-ideas page only — no answers.

> 本页给老师看（含答案），不要发给学生。学生版只用 Student PDF。